

TrES-3b

R-1030

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_161004 · created 2026-07-30T06:09:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457665.5893 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.043
Transit depth (Rp/Rs)^2	3.86 +/- 1.71 [%]
Orbital Inclination (inc)	81.53 +/- 1.37
Airmass coefficient 1 (a1)	1.055 +/- 0.013
Airmass coefficient 2 (a2)	-0.04 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#3 - [477, 338]
Optimal Aperture	3.3
Optimal Annulus	7.22
Transit Duration (day)	0.0278 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.043 vs 0.16309 (deviation 0.51σ)
Duration fit vs published	0.0278 d vs 0.0567 d (deviation 2.14σ)
Mid-time O–C	-3.8 min
Depth SNR	3.0
Residual scatter	1.26 %

Light curve

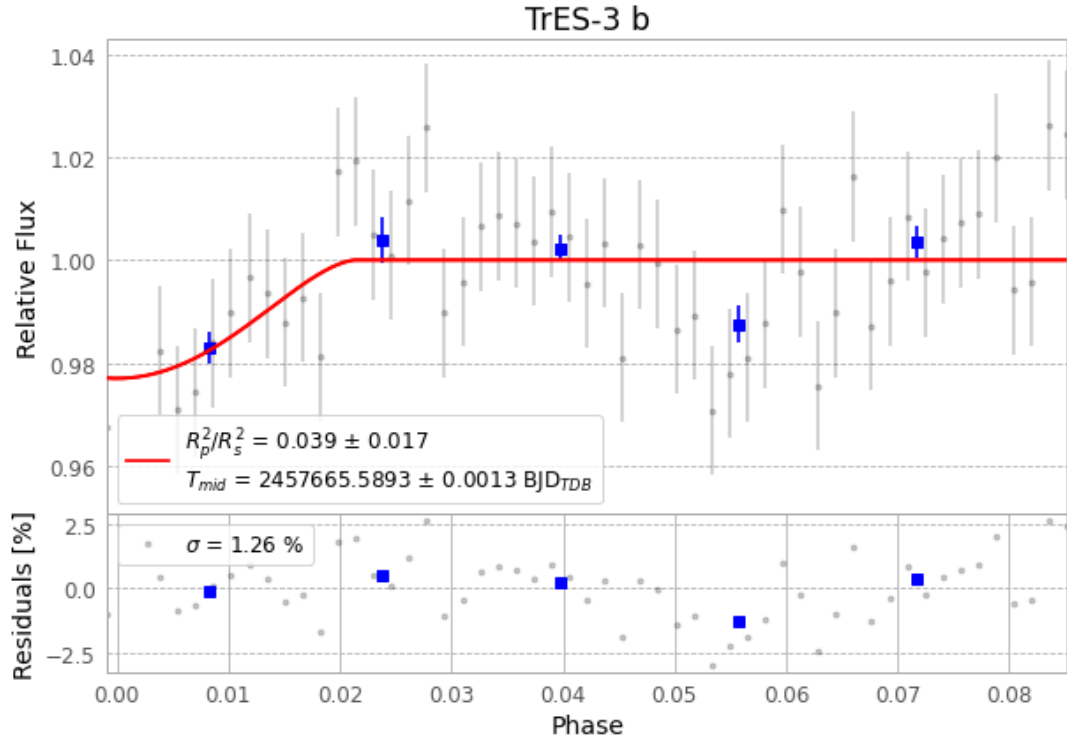


Figure 1 — FinalLightCurve_TrES-3 b_2016-10-03.png (SHA-256 579332b2dadca7f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [156, 255], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0b28044e888c759a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d840746

Data provenance

53 FITS frames (35 MB), TRES-3161004020615.FITS → TRES-3161004044812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-161004.log (f1a99c2a342d...), FinalLightCurve_TrES-3 b_2016-10-03.png (579332b2dadca...), FinalLightCurve_TrES-3 b_2016-10-03.csv (538fb7c8850c...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 06:09Z.